

Section: Particles and Waves

Topic: The Standard Model

The diagram shows some particles involved in the decay of a neutron.

(a)

Name the fundamental force responsible for this decay

✓ The **weak nuclear force** is responsible for beta decay.

(b)

Identify the missing particle in the decay equation

In beta decay:

neutron $\rightarrow$ proton + electron + ?

✓ The missing particle is the **electron antineutrino** ($\bar{\nu}_e$).

(c)

State a property of the missing particle that makes it difficult to detect

✓ Electron antineutrinos have **very little mass** (almost zero) and **no electric charge**, so they interact **very weakly** with matter, making them extremely difficult to detect.

(d)

Neutrons and protons are baryons.

What is meant by a baryon?

✓ A **baryon** is a particle composed of **three quarks**.

Examples: protons (uud) and neutrons (udd).

 Revision Tips

- **Beta decay** involves a **down quark turning into an up quark**, mediated by the **weak force**
- Charge and lepton number must be **conserved**
- The **antineutrino** ensures conservation of lepton number
- Neutrinos are notoriously difficult to detect — they pass through ordinary matter almost unaffected
- Baryons: 3 quarks; Mesons: 1 quark + 1 antiquark